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Does Central Bank Tone Matter for Fama-French Factor Returns?

AFMLC 2025

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Central bank communication, particularly **speeches**, has emerged as a crucial monetary policy tool, especially after the **2008-2009 global financial crisis**.



The **tone and context** in which words are expressed have a predictive effect on the future path of monetary policy and on financial market developments. As Ben Bernanke noted:

> *"Monetary policy is 98 percent talk and only two percent action."*

Understanding **how market participants perceive and respond to central bank communication** is essential for both academic research and investment practice.

> *Research Question*: **Does the tone of Federal Reserve speeches have a predictive impact on Fama-French five factors and the momentum factor?**

Conclusion & Implications

The tone of Federal Reserve speeches significantly influences factor returns , confirming that market participants closely follow central bank communication. Sentiment indicators specific to monetary policy contexts (ABG and LLM) demonstrated the strongest predictive power.

For Investors

The tone of central bank communication can serve as a **valuable input for investment strategies**. Hawkish tone signals potential underperformance of momentum, growth, and aggressive investment portfolios.

For Policymakers

Highlights the critical importance of careful and precise communication. Both intended and unintended **impacts** of words in speeches can significantly affect financial asset pricing and market expectations.

For Academia

Supports the behavioral finance view that investor behavior is not always fully rational. Sentiment plays a meaningful role in asset pricing, **challenging the semi-strong form of market efficiency**.



As central bank speeches become more hawkish, **value stocks and conservative investment portfolios** tend to outperform growth stocks and aggressive investment portfolios, respectively. The **Apel-Grimald sentiment index** closely mirrors LLM-generated sentiment, offering a transparent and reproducible alternative.

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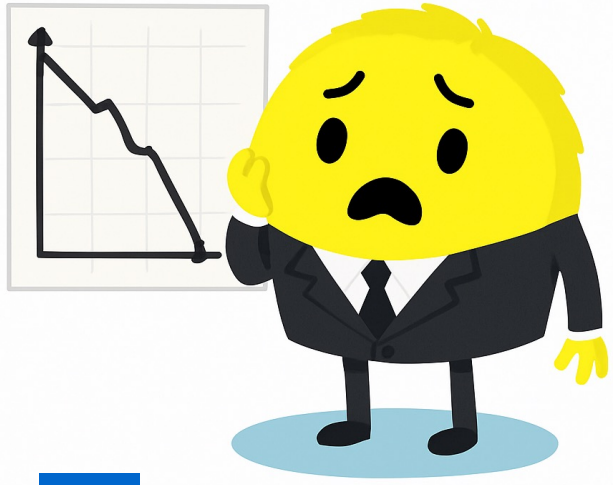


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Evolution of Central Bank Communication

Prior to the 1990s, central banks were characterized by low transparency and cryptic statements. Over the past three decades, there has been a remarkable shift toward transparent communication to shape market expectations and guide monetary policy.

Wecken & Büchsenmann (2019); De Haan et al. (2007)



Communication at the Zero Lower Bound

After the 2008-2009 financial crisis, many central banks reached the zero lower bound (ZLB) of interest rates. Communication emerged to fill the role of strategically shaping market expectations, becoming a crucial monetary policy tool when traditional instruments were constrained.

Woodford (2005); Hansen & McMahon (2016)

Sentiment Analysis & Market Efficiency

Sentiment analysis quantifies the tone of financial texts. The behavioral literature argues for imperfect information processing by market participants, challenging the efficient market hypothesis. Several studies found that central bank sentiment has a significant impact on equity returns.

Schmeling & Wagner (2015); Sadique et al. (2013); Hong et al. (2007)

> **Semi-strong form of market efficiency** *(Fama, 1970):* it is not possible to predict the future evolution of profitable factors returns by extracting information fromcentral bank communication.

> Only **unexpected announcements, statements** or tone can have significant effect on the financial market.



> **Behavioural literature** argues for the imperfect information processing of market participants.

> **Gradual-information-diffusion** hypothesis states information only diffuses with a **certain lag stemming** from limited information-processing capacity of investors.

Assessing the tone of central bank speeches and their relation to factor returns can help to understand the investors' bounded rationality and the validity of the efficient market hypothesis.

Doubt in efficient market hypothesis

Several find that market participants are not fully rational and therefore irrational behaviours can lead to mispricing of assets on the financial market

Andersen et al., 2007; Savor and Wilson, 2013; Kurov et al., 2016; Hu et al., 2017



Several study conclude central bank communication influences financial markets

ECB tone has a statistically and economically significant effect on equity and government bond prices on the day of the press conference. Federal Reserve Beige Book and FOMC minutes tone drives volatility, return and trading volume in the stock market.

Schmeling and Wagner (2015); Sadique et al. (2013); Jegadeesh and Wu (2017)

... However some argue that this is not always the case

Study focuses on whether the tone of ECB communication influences stock returns in the Euro area. Based on the findings, the effect of such announcements on stock returns has not been significant.

Kurihara (2014)

Two types of dictionaries based on one-word expressions

- *Loughran-Mcdonald dictionary (2012)* is one of the most widely used methods for sentiment analysis of financial text. LM dictionary might lack explanatory power in a monetary policy context.
- *Bennani and Neuenkirch (2017)* sorted keywords into two categories depending on whether they indicate a more hawkish tone (i.e. increase, high) or a more dovish tone (low, decrease).

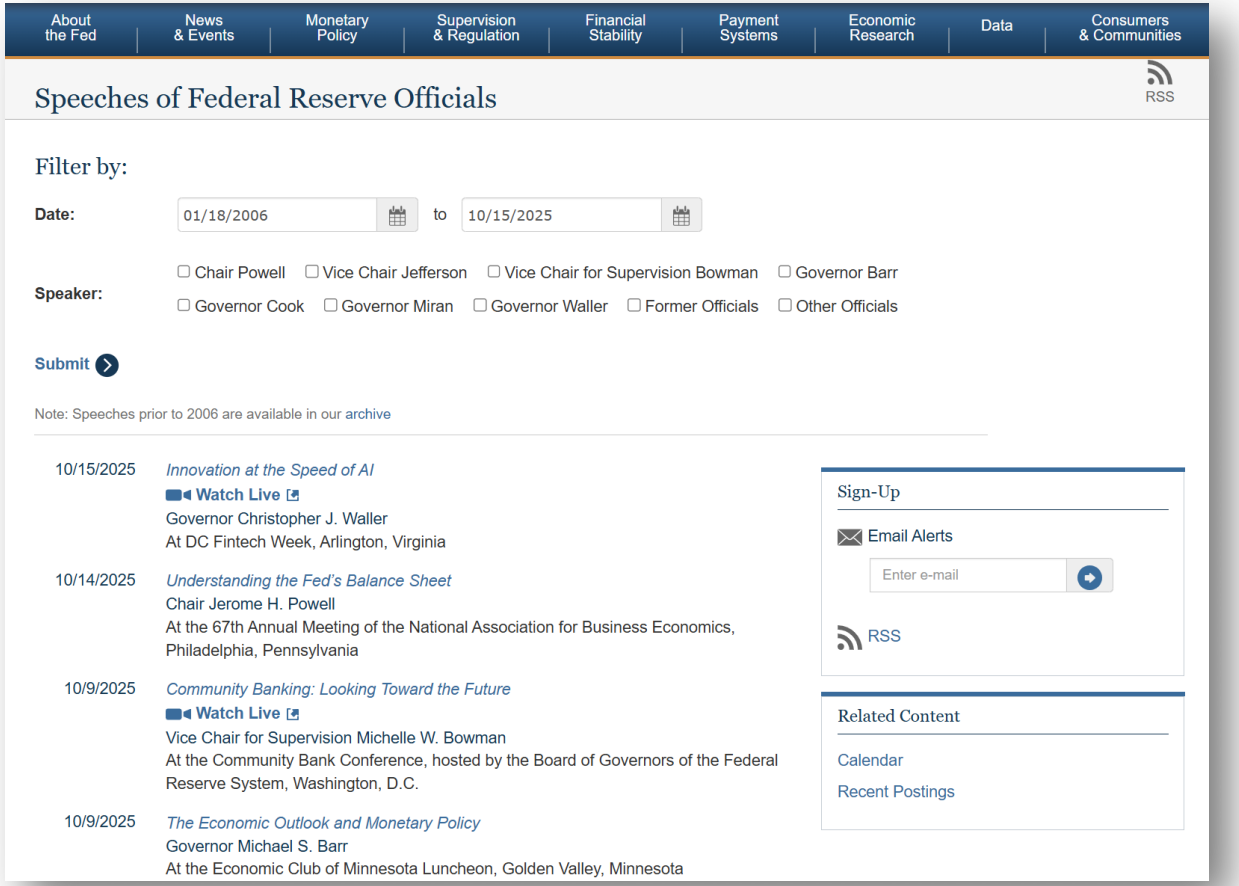
Methods capturing combinations of words

Apel-Grimaldi (2012)			*Own dictionary*		
inflation, cyclical position, growth, price, wages, oil price, development	high*, strong*, increas*, fast	low*, weak*, decreas*, slow	gdp, output, inflation, price, employment	accelerat*, bettter, boom*, emerg*, expansion, fast*, favo(u)rabl*, firm*, great*, high*, improv*, increas*, larger, positive, rais*, ris*, stabili*, stable, strengthen*, strong*, subdued, unsustainable, upside, upswing, upturn, upward*.	collaps*, contraction, dampen*, decelerat*, declin*, decreas*, delay*, depression, destabili*, deteriorat*, difficul*, diminish*, disappear*, downside, downswing, downturn, downward*, fall*, fragil*, low*, negative, poor, recession*, slow*, sluggish, small*, struggling, sustainable, unfavo(u)rabl*, unstable, weak*, worse





We applied webscraping to retrieve all the speeches from 2006 to the end of 2023 and distinguish the speaker of the speech.



Year	Number of speeches	Share of Chairmen and Chairs
2006	73	33%
2007	73	32%
2008	72	32%
2009	54	39%
2010	60	43%
2011	48	38%
2012	42	38%
2013	53	32%
2014	41	37%
2015	54	20%
2016	43	12%
2017	55	24%
2018	44	25%
2019	78	22%
2020	53	11%
2021	68	13%
2022	46	13%
2023	95	9%

```
prompt = """Determine whether the central bank statement reflects a hawkish or dovish stance,
considering inflation, economic output, growth, employment, monetary policy tools, cyclical position
and global economic conditions. Evaluate how relevant the text is to a central bank's core monetary
policy decisions such as interest rate announcements or forward guidance, versus unrelated or
economically insignificant events (e.g., charity appearances). Focus on the event's context and
market message using a scale from 1 (irrelevant) to 100 (highly relevant). You can answer with None
if you can't determine the sentiment or relevance.

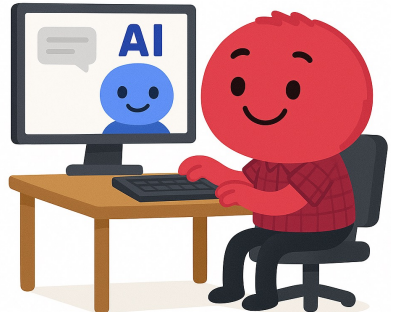
Example:
"The Federal Reserve's decision to raise interest rates by 25 basis points is a clear indication of
its commitment to combating inflation." > hawkish_sentiment: 100, relevance: 100

"The central bank is monitoring the economic situation closely and worried about unexpectedly high
unemployment data." > hawkish_sentiment: 20, relevance: 80

"During the 1930s, the Fed's policies were influenced by the Great Depression, leading to a focus on
economic recovery." > hawkish_sentiment: 10, relevance: 10
"""
```



Gemma 3



LLM vs. Traditional Methods

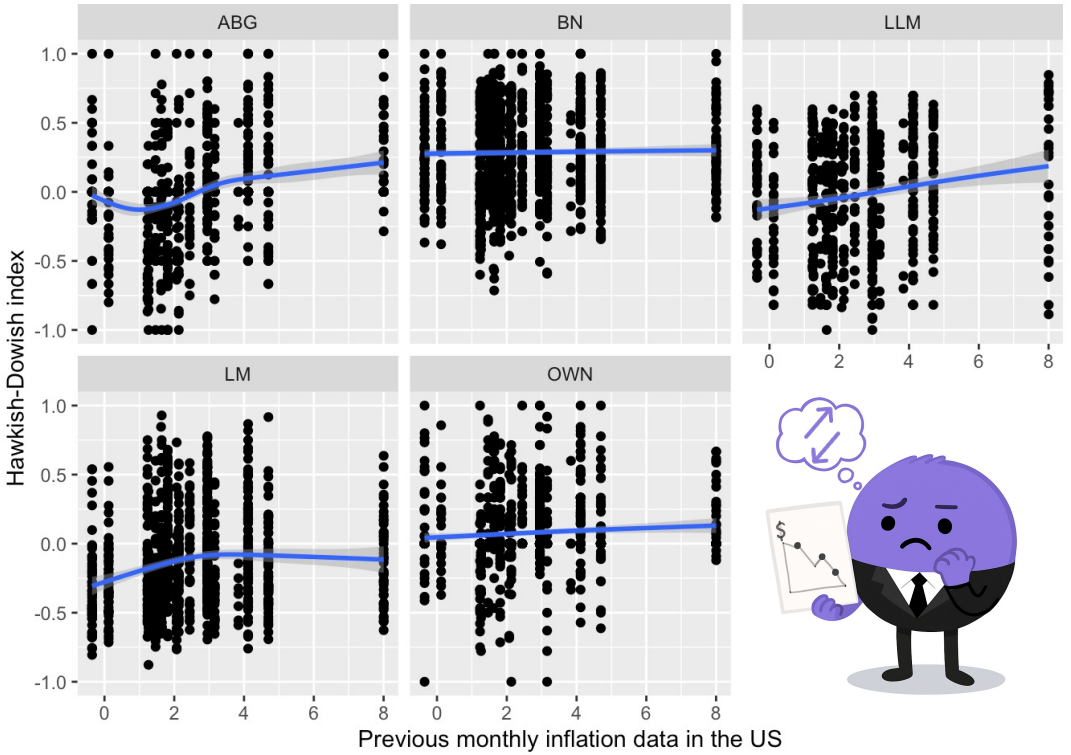
The Apel-Grimaldi (ABG) sentiment index closely mirrors the LLM-generated sentiment scores, demonstrating strong convergent validity.

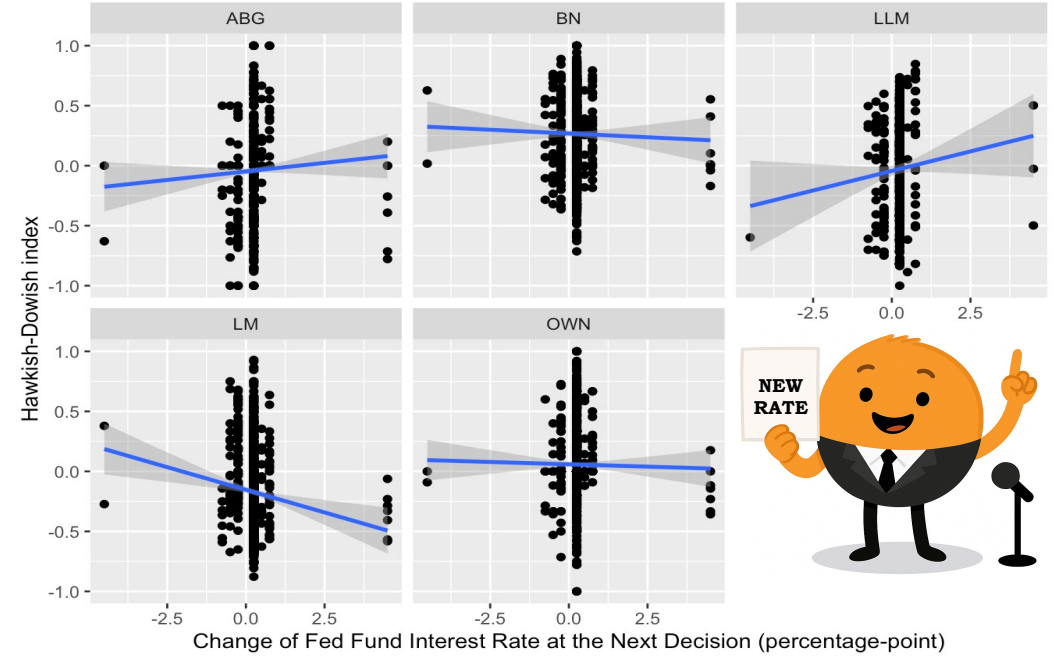
Both ABG and LLM indices correlate with inflation trends and Federal Funds rate changes , confirming their ability to capture monetary policy stance.

Both methods aid in market prediction, with the LLM showing marginally superior consistency during deflationary periods.

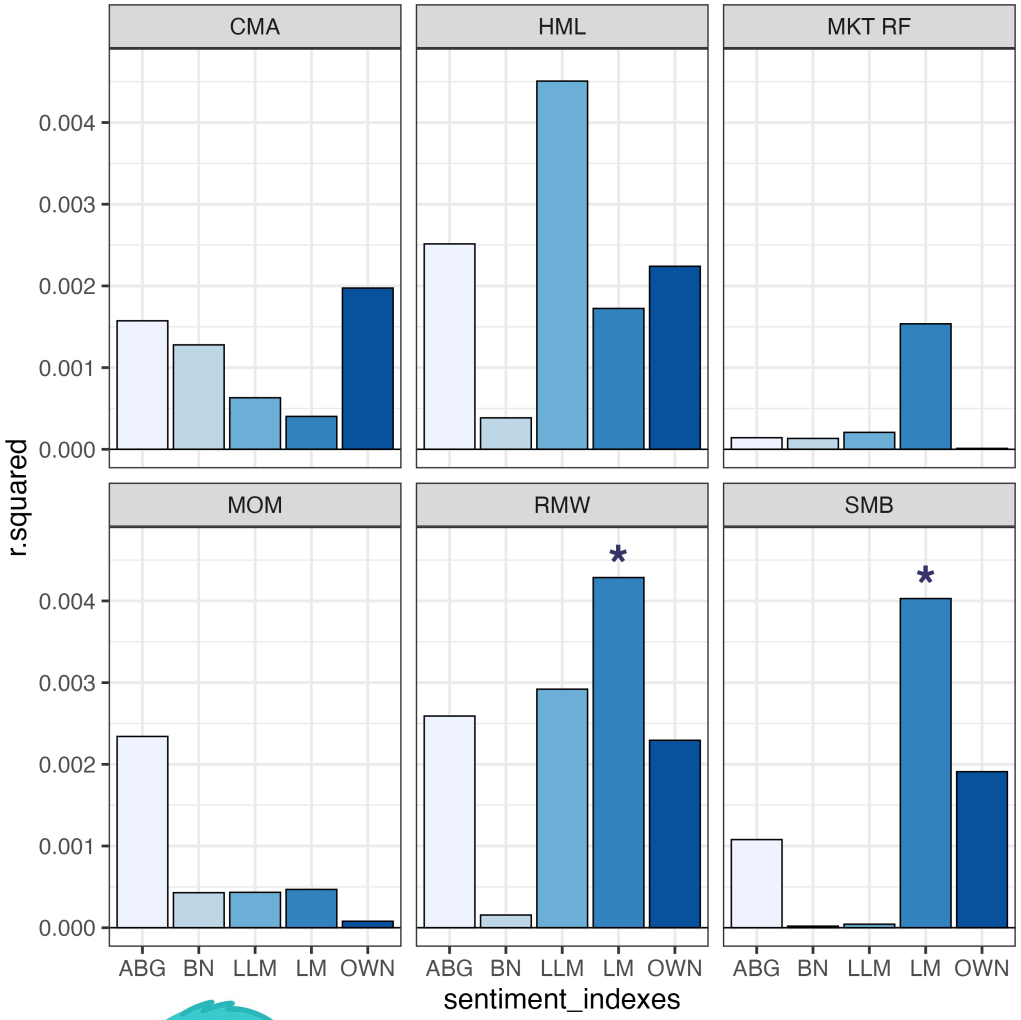
The difference in performance between LLM and ABG methods is minor, suggesting that dictionary-based approaches might still be usefull.

Key Advantage: The ABG method offers a reproducible and transparent index, in contrast to the black-box nature of LLM applications.

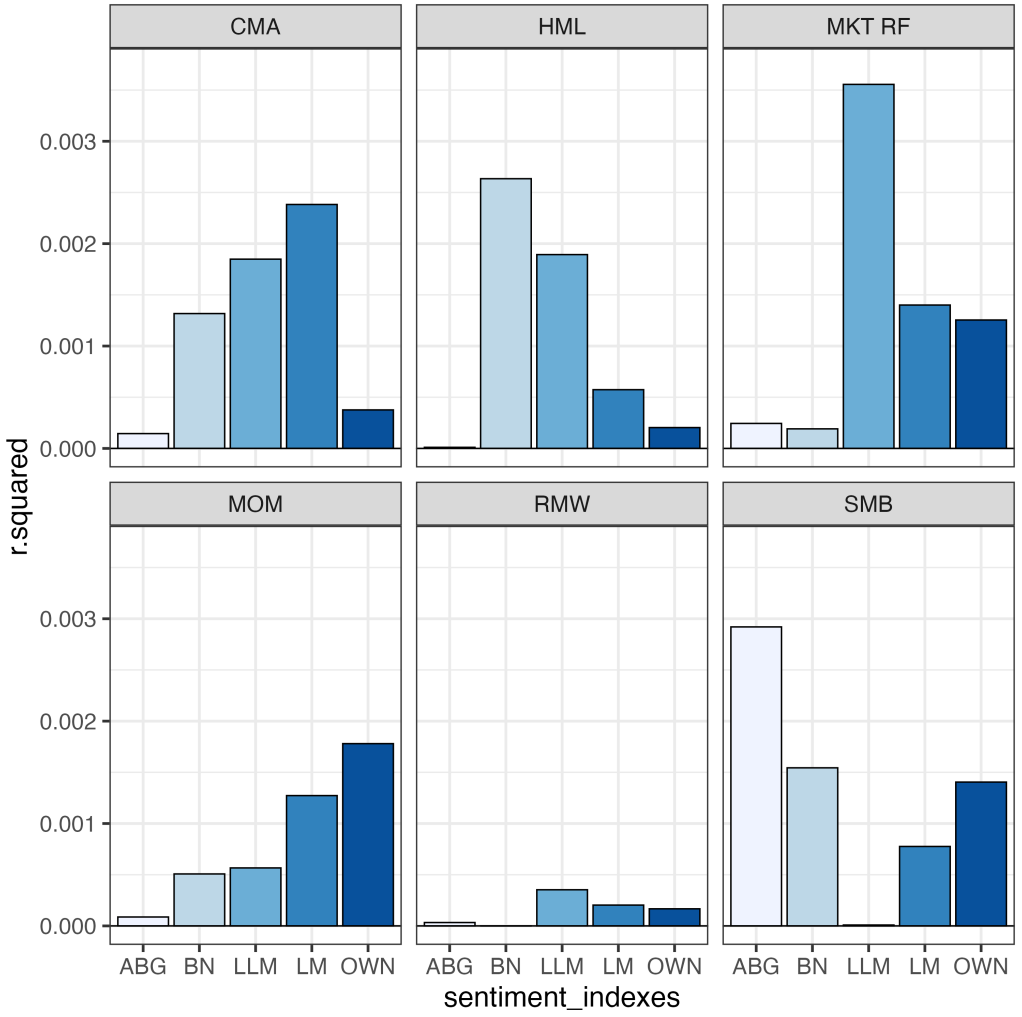




~ Market Factors$_t$



~ Market Factors$_{t+1}$



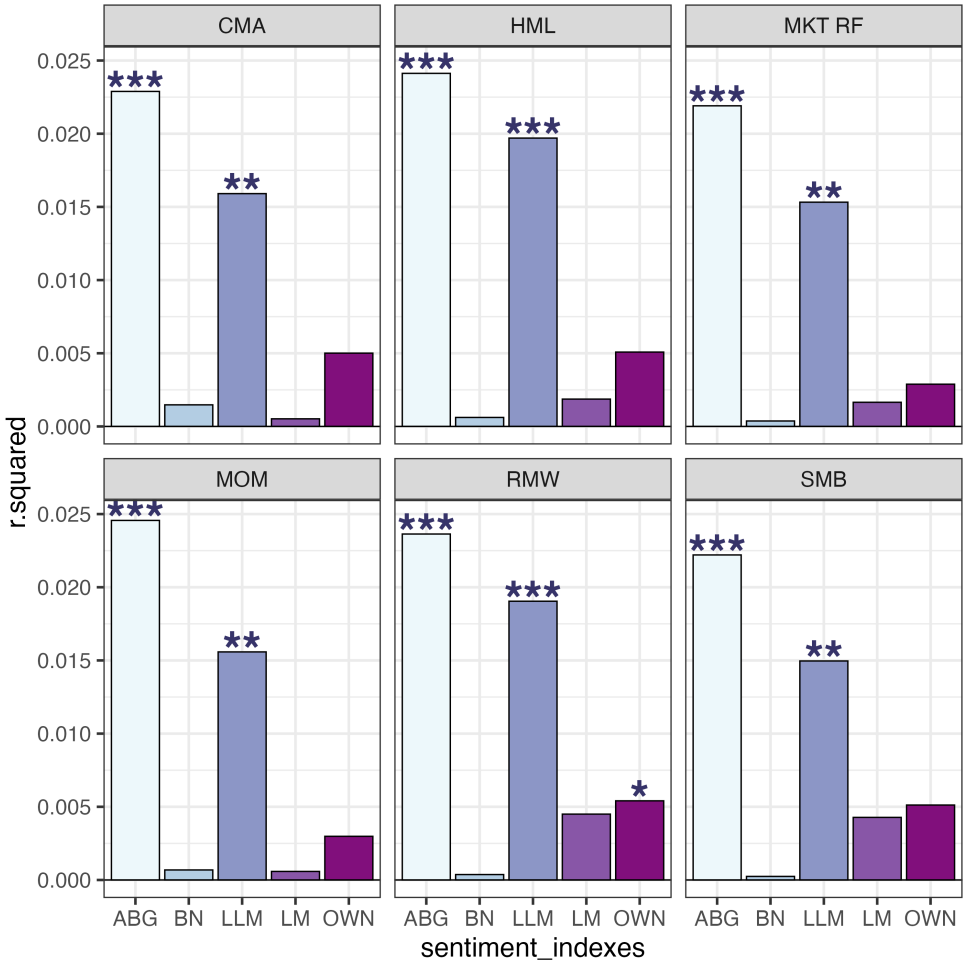


Bivariate OLS results

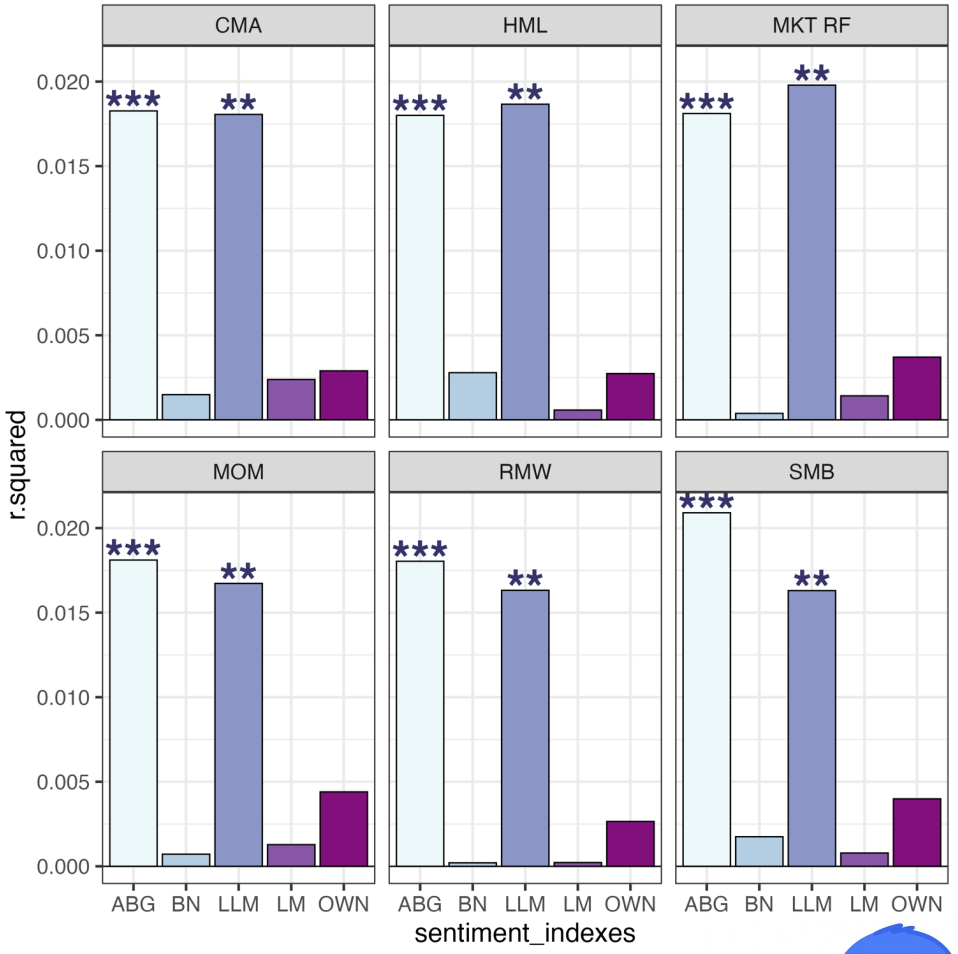
*** p < 0.01, ** p < 0.05, * p < 0.1

Filtering to cases when speech was delivered by a member of the Chair

~ Market Factors$_t$



~ Market Factors$_{t+1}$



We controlled for whether the speech was delivered by the **Chairman or Chair of the Federal Reserve** to account for potential differences in market impact based on speaker authority.



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